

Exercise 1a

Set $|1\rangle$ to have 0 energy. Move to rotating frame, valid if $\delta(t)$ assumed to vary slowly compared to ..., apply RWA to get

$$\mathcal{H} = -\hbar\delta(t)|2\rangle\langle 2| - \hbar\left(\frac{\Omega}{2}|2\rangle\langle 1| + \frac{\Omega}{2}|1\rangle\langle 2|\right) \simeq -\hbar\begin{pmatrix} 0 & \Omega/2 \\ \Omega/2 & \delta(t) \end{pmatrix}$$

Eigenvalues E of matrix given by

$$-E(\delta(t) - E) - \frac{\Omega^2}{4} = 0 \implies E = -\frac{\hbar}{2}(\delta \pm \underbrace{\sqrt{\delta^2 + \Omega^2}}_{:=\Omega_R})$$

With eigenvectors satisfying

$$E \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} -\hbar\Omega/2v_2 \\ -\hbar\Omega/2v_1 - \hbar\delta(t)v_2 \end{pmatrix}$$

This gives us $v_2 = -\frac{E}{\hbar}v_1$. Let E_A, E_B refer to the $+, -$ eigenstates respectively. Then, our corresponding eigenstates are

$$\begin{aligned} E_A &= -\frac{\hbar}{2}(\delta + \Omega_R), & |A(t)\rangle &= \frac{1}{N_A}(\Omega|1\rangle + (\delta + \Omega_R)|2\rangle) \\ E_B &= -\frac{\hbar}{2}(\delta - \Omega_R), & |B(t)\rangle &= \frac{1}{N_B}(\Omega|1\rangle + (\delta - \Omega_R)|2\rangle) \end{aligned}$$

with the normalizations given by

$$N_A = \sqrt{\Omega^2 + (\delta + \Omega_R)^2}, \quad N_B = \sqrt{\Omega^2 + (\delta - \Omega_R)^2}$$

As $\delta \rightarrow \infty$, $\Omega_R \rightarrow \delta$ and we get

$$\begin{aligned} E_A &\rightarrow -\hbar\delta, & |A(t)\rangle &\rightarrow |2\rangle \\ E_B &\rightarrow 0, & |B(t)\rangle &\rightarrow |1\rangle \end{aligned}$$

As $\delta \rightarrow -\infty$, $\Omega_R \rightarrow -\delta$ and

$$\begin{aligned} E_A &\rightarrow 0, & |A(t)\rangle &\rightarrow |1\rangle \\ E_B &\rightarrow -\hbar\delta, & |B(t)\rangle &\rightarrow |2\rangle \end{aligned}$$

which is expected as $|\delta| \gg \Omega$ implies a weak driving field. In the limit $\Omega = 0$, our hamiltonian is diagonal with $|1\rangle$ corresponding to eigenenergy 0 and $|2\rangle$ to $-\hbar\delta = \hbar(\omega - \nu)$ (the energy of $|2\rangle$ in the non-rotating frame shifted down by ν). See the Figure below for reference.

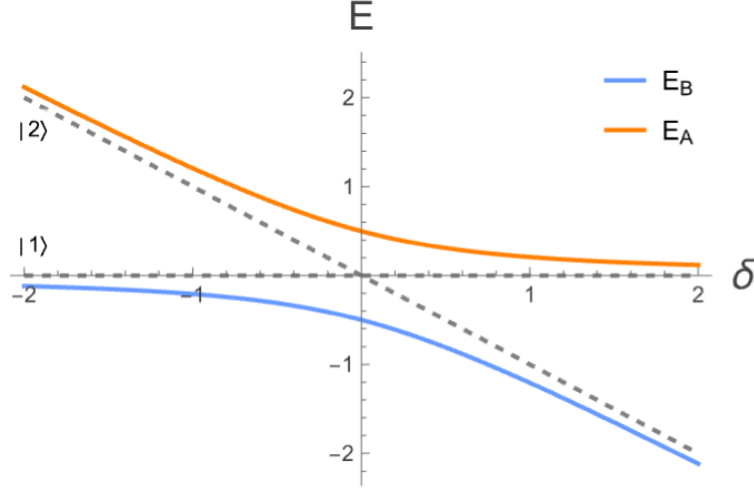


Figure 1: Energy of dressed states against detuning for 2 level system in Problem 1 with dashed lines indicating energies for zero field eigenstates of 0 and $-\hbar\delta$

Exercise 1b

Remark: $i\hbar|\dot{A}(t)\rangle \neq H|A(t)\rangle$ because $A(t)$ is an instantaneous eigenvector of H and the time dependence is due only to $\delta(t)$.

With $|\psi(t)\rangle = a(t)|A(t)\rangle + b(t)|B(t)\rangle$, we get

$$i\hbar|\dot{\psi}(t)\rangle = i\hbar\left(\dot{a}(t)|A(t)\rangle + a(t)|\dot{A}(t)\rangle + \dot{b}(t)|B(t)\rangle + b(t)|\dot{B}(t)\rangle\right)$$

and

$$H|\psi(t)\rangle = a(t)E_A|A(t)\rangle + b(t)E_B|B(t)\rangle$$

Thus, applying $\langle A(t)|$ and $\langle B(t)|$ to both sides of the schrodinger equation gives

$$i\hbar\dot{a} + i\hbar b\langle A|\dot{B}\rangle = aE_A$$

$$i\hbar a\langle B|\dot{A}\rangle + i\hbar\dot{b} = bE_B$$

where $\langle A|A\rangle = \langle B|B\rangle = 1$ due to normalization and $\langle A|\dot{B}\rangle = \langle B|\dot{A}\rangle = 0$ by Lemma 1. Thus, it remains to calculate $\langle A|\dot{B}\rangle$. We leave this to mathematica (see the accompanying notebook). Our result is

$$\langle A|\dot{B}\rangle = \frac{\dot{\delta}\Omega}{2(\delta^2 + \Omega^2)} = -\langle B|\dot{A}\rangle$$

Plugging in these inner products gives us the EOMs

$$i\hbar\dot{a} + i\hbar b \frac{\dot{\delta}\Omega}{2(\delta^2 + \Omega^2)} = aE_A$$

$$-i\hbar a \frac{\dot{\delta}\Omega}{2(\delta^2 + \Omega^2)} + i\hbar\dot{b} = bE_B$$

Simplifying gives us

$$\dot{a} = -\frac{iE_A}{\hbar}a - \frac{\dot{\delta}\Omega}{2(\delta^2 + \Omega^2)}b \quad (1)$$

$$\dot{b} = -\frac{iE_B}{\hbar}b + \frac{\dot{\delta}\Omega}{2(\delta^2 + \Omega^2)}a \quad (2)$$

In the adiabatic limit, $\dot{\delta} \approx 0$ as it changes very slowly, we get

$$\dot{a} = -\frac{iE_A}{\hbar}a, \quad \dot{b} = -\frac{iE_B}{\hbar}b$$

this implies that our state $|\psi(t)\rangle$ just picks up phase factors on the basis states $|A(t)\rangle, |B(t)\rangle$ and does not mix these states. We can then, for example, follow $|A(t)\rangle$ from $\delta \ll 0$ to $\delta \gg 0$ to traverse from $|2\rangle \rightarrow |1\rangle$.

Lemma 1. *We claim the following results*

1. *If $|A(t)\rangle$ is normalized and exists in a real vector space, then $\langle A|\dot{A}\rangle = 0$*
2. *Eigenvectors of Hermitian matrix are orthogonal*

Proof of 1. By assumption, $\langle A(t)|A(t)\rangle = 1$, so

$$0 = \frac{d}{dt}\langle A(t)|A(t)\rangle = \langle \dot{A}|A\rangle + \langle A|\dot{A}\rangle = \langle A|\dot{A}\rangle^* + \langle A|\dot{A}\rangle = 2\langle A|\dot{A}\rangle$$

□

Proof of 2. Let v_i be eigenvectors of hermitian operator, A , with different eigenvalues, λ_i . Then

$$\langle v_i|A|v_j\rangle = \lambda_i\langle v_i|v_j\rangle = \lambda_j\langle v_i|v_j\rangle \implies (\lambda_i - \lambda_j)\langle v_i|v_j\rangle = 0$$

□

Exercise 1c

We have $a(t) = a(0)e^{-iE_A t/\hbar}$ and $b(t) = b(0)e^{-iE_B t/\hbar}$. For $|\psi(0)\rangle = |1\rangle = a(0)|A(0)\rangle + b(0)|B(0)\rangle \approx a(0)|2\rangle + b(0)|1\rangle$, if we begin very negatively detuned. This gives $a(0) = 0$ and $b(0) = 1$. Thus, $a(t) = 0$ and $b(t) = e^{-iE_B t/\hbar}$. Hence,

$$|\psi(t)\rangle = e^{-iE_B t/\hbar}|B(t)\rangle$$

Thus, if we adiabatically change the detuning from negative to positive, $|B(t)\rangle$ goes from state $|1\rangle \rightarrow |2\rangle$ as shown in 1a. Thus, we can take the initial population in state 1 to state 2.

Exercise 1d

We want equations 1 and 2 to be uncoupled. The coupling term is $\dot{\delta}\Omega/(2(\delta^2 + \Omega^2))$. The other relevant physical quantity is the energy *difference* $|E_A - E_B|/\hbar$. For adiabatic passage, we thus need

$$\frac{\dot{\delta}\Omega}{2(\delta^2 + \Omega^2)} \ll \frac{1}{\hbar}(E_A - E_B) = \frac{1}{2}(\delta + \Omega_R - (\delta - \Omega_R)) = \Omega_R$$

Thus,

$$\dot{\delta} \ll \frac{2\Omega_R(\delta^2 + \Omega^2)}{\Omega}$$

The lower limit of this expression occurs when $\delta = 0$ and $\Omega_R \rightarrow \Omega$. Thus, for the inequality to hold at all δ , we require

$$\dot{\delta} \ll \frac{2\Omega(\Omega^2)}{\Omega} = 2\Omega^2$$

as desired.

For a resonant π pulse, we use a pulse duration τ defined by $\Omega T \tau = \pi \implies \tau = \pi/\Omega$. In comparison, if $\frac{d\delta}{dt} \ll 2\Omega^2$, then we have

$$\delta(T) - \delta(0) \ll 2\Omega^2 T \implies T \gg \frac{\delta(T) - \delta(0)}{2\Omega^2} > \frac{2\Omega}{\Omega^2} = \frac{1}{\Omega} \approx \frac{\pi}{\Omega} = \tau$$

where population inversion requires $|\delta| \gg \Omega$ and if $\delta(0) < 0$, then $\delta(T) - \delta(0) > 2\Omega$. We conclude that adiabatic passage thus requires much longer times than resonant pulses.

Exercise 1e**Advantages**

1. No need to stay on resonance during entire measurement period
2. Detuning is easily controlled by frequency of driving field.

Disadvantages

1. Takes longer than pulses
2. Pulsed measurement schemes common in other fields like magnetic resonance.
More literature/infrastructure to borrow from

Exercise 2a

We work in the rotating frame with energies shifted down by ν so that $|2\rangle$ has energy $\hbar(\omega_{12} - \nu) = -\hbar\delta$ and $|3\rangle$ has energy $\hbar((\omega_{12} + \Delta\omega) - \nu) = \hbar(\Delta\omega - \delta)$. Then, our hamiltonian in the rotating frame with the RWA takes the form

$$\mathcal{H} = -\hbar \begin{pmatrix} 0 & \Omega/2 & \Omega/2 \\ \Omega/2 & \delta & 0 \\ \Omega/2 & 0 & -(\Delta\omega - \delta) \end{pmatrix}$$

as expressed in the basis $\{|1\rangle, |2\rangle, |3\rangle\}$. We also took Ω to be real by appropriate choice of phase factor for the basis states. Writing an arbitrary state as $|\psi(t)\rangle = c_1(t)|1\rangle + c_2(t)|2\rangle + c_3(t)|3\rangle$, the schrodinger equation gives us $i\hbar \frac{d}{dt}|\psi(t)\rangle = \mathcal{H}|\psi(t)\rangle$ so that

$$\begin{aligned} \dot{c}_1(t)|1\rangle + \dot{c}_2(t)|2\rangle + \dot{c}_3(t)|3\rangle &= ic_1 \frac{\Omega}{2}|2\rangle + ic_1 \frac{\Omega}{2}|3\rangle + ic_2 \frac{\Omega}{2}|1\rangle \\ &\quad + ic_2\delta|2\rangle + ic_3 \frac{\Omega}{2}|1\rangle - ic_3(\Delta\omega - \delta)|3\rangle \end{aligned}$$

Orthogonality gives us the equations

$$\begin{aligned} \dot{c}_1 &= i\frac{\Omega}{2}c_2 + i\frac{\Omega}{2}c_3 \\ \dot{c}_2 &= i\frac{\Omega}{2}c_1 + i\delta c_2 \\ \dot{c}_3 &= i\frac{\Omega}{2}c_1 - i(\Delta\omega - \delta)c_3 \end{aligned}$$

Exercise 2b

We use adiabatic expansion to deal with c_3 . Our first order expansion, $\dot{c}_3^{(1)}$ comes from setting $\dot{c}_3 = 0$, so that

$$0 = \dot{c}_3 = i\frac{\Omega}{2}c_1 - i(\Delta\omega - \delta)c_3^{(1)} \implies c_3^{(1)} = \frac{\Omega}{2(\Delta\omega - \delta)}c_1$$

Our second order expansion $c_3^{(2)}$ satisfies

$$\begin{aligned} \dot{c}_3^{(1)} &= i\frac{\Omega}{2}c_1 - i(\Delta\omega - \delta)c_3^{(2)} \\ \frac{\Omega}{2(\Delta\omega - \delta)}\dot{c}_1 &= i\frac{\Omega}{2}c_1 - i(\Delta\omega - \delta)c_3^{(2)} \\ \frac{\Omega}{2(\Delta\omega - \delta)}\left(i\frac{\Omega}{2}c_2 + i\frac{\Omega}{2}c_3^{(2)}\right) &= i\frac{\Omega}{2}c_1 - i(\Delta\omega - \delta)c_3^{(2)} \end{aligned}$$

From this, we get

$$\begin{aligned}\Omega^2 c_2 + \Omega^2 c_3^{(2)} &= 2\Omega(\Delta\omega - \delta)c_1 - 4(\Delta\omega - \delta)^2 c_3^{(2)} \\ c_3^{(2)}(\Omega^2 + 4(\Delta\omega - \delta)^2) &= 2\Omega(\Delta\omega - \delta)c_1 - \Omega^2 c_2 \\ c_3^{(2)} &= \frac{2\Omega(\Delta\omega - \delta)c_1 - \Omega^2 c_2}{\Omega^2 + 4(\Delta\omega - \delta)^2}\end{aligned}$$

Plugging back into our original equation for $\dot{c}_1$, we get

$$\begin{aligned}\dot{c}_1 &= i\frac{\Omega}{2}c_2 + i\frac{\Omega}{2}\left(\frac{2\Omega(\Delta\omega - \delta)c_1 - \Omega^2 c_2}{\Omega^2 + 4(\Delta\omega - \delta)^2}\right) \\ &= \frac{i\Omega}{2}\left(1 - \frac{\Omega^2}{\Omega^2 + 4(\Delta\omega - \delta)^2}\right)c_2 + \frac{i\Omega^2(\Delta\omega - \delta)}{\Omega^2 + 4(\Delta\omega - \delta)^2}c_1\end{aligned}$$

Since we are expanding to second order, $\Omega^2/(\Delta\omega - \delta)^2 \ll 1 \implies \Omega^2 \ll (\Delta\omega - \delta)^2$. With this in mind, we get

$$\begin{aligned}\dot{c}_1 &\approx \frac{i\Omega}{2}\left(1 - \frac{\Omega^2}{4(\Delta\omega - \delta)^2}\right)c_2 + \frac{i\Omega^2(\Delta\omega - \delta)}{4(\Delta\omega - \delta)^2}c_1 \\ &= \frac{i\Omega}{2}\left(1 - \frac{\Omega^2}{4(\Delta\omega - \delta)^2}\right)c_2 + \frac{i\Omega^2}{4(\Delta\omega - \delta)}c_1\end{aligned}$$

Exercise 2c

We use our first order adiabatic expansion, $c_3(1) = \frac{\Omega}{2(\Delta\omega - \delta)}c_1$ from part (b) to get the system

$$\begin{aligned}\dot{c}_1 &= \frac{i\Omega}{2}c_2 + \frac{i\Omega^2}{4(\Delta\omega - \delta)}c_1 \\ \dot{c}_2 &= \frac{i\Omega}{2}c_1 + i\delta c_2\end{aligned}$$

If the effects of state $|3\rangle$ were ignored ($\Delta\omega \rightarrow \infty$), we would have Rabi oscillations at frequency $\Omega/2$ during resonance. To achieve oscillations, we require $\ddot{c}_i \propto -c_i$ for $i = 1, 2$. Differentiating c_1 twice, we see that

$$\begin{aligned}\ddot{c}_1 &= \frac{i\Omega}{2}\left(\frac{i\Omega}{2}c_1 + i\delta c_2\right) + \frac{i\Omega^2}{4(\Delta\omega - \delta)}\left(\frac{i\Omega}{2}c_2 + \frac{i\Omega^2}{4(\Delta\omega - \delta)}c_1\right) \\ &= -\left(\frac{\Omega^2}{4} + \frac{\Omega^4}{16(\Delta\omega - \delta)^2}\right)c_1 + \frac{i\Omega}{2}\left(i\delta + \frac{i\Omega^2}{4(\Delta\omega - \delta)}\right)c_2\end{aligned}$$

Thus, we get the desired form for $\ddot{c}_1$ iff

$$\delta = -\frac{\Omega^2}{4(\Delta\omega - \delta)} \implies \delta = 2\Delta\omega \pm \sqrt{4\Delta\omega^2 + \Omega^2}$$

Take the + solution since the problem assumes $\delta = \nu - \omega_{12} > 0$.

If such δ is chosen, then the effect of state 3 is to induce a phase shift so that the oscillation frequency goes from $\frac{\Omega}{2} \rightarrow \sqrt{\frac{\Omega^2}{4} + \delta^2}$. Otherwise, for relatively weak driving fields or large detuning, our system looks like $\dot{c}_1 = -i\delta c_1$ and $\dot{c}_2 = o\delta c_2$, so that state 3 causes energy shifts by $\delta = -\Omega^2/(4(\delta\omega - \delta))$. This looks like an AC stark shift.

Exercise 2d

Exercise 3a

We expand $|\psi(t)\rangle$ in terms of the bound state $|b\rangle$ and continuum states, indexed by $|k\rangle$, as

$$|\psi(t)\rangle = b(t)|b\rangle + \sum_{k=-\infty}^{\infty} c_k(t)|k\rangle$$

Schrödinger's equation gives us

$$\begin{aligned} i\hbar\dot{b} &= \langle b|\mathcal{H}|\psi(t)\rangle = -\hbar \sum_{k=-\infty}^{\infty} f^* g c_k \\ i\hbar\dot{c}_k &= \langle k|\mathcal{H}|\psi(t)\rangle = \hbar(\omega_k - \nu)c_k - \hbar f g b \end{aligned}$$

Set $\delta_k = \nu - \omega_k$. Thus, we get the system

$$\begin{aligned} \dot{b} &= i \sum f^* g c_k \\ \dot{c}_k &= i\delta_k c_k + i f g b \end{aligned}$$

Formally integrating c_k gives us

$$c_k(t) = i g \int_0^t e^{i\delta_k(t-\tau)} f(\tau) b(\tau) d\tau + c_k(0) e^{i\delta_k t}$$

We assume that the $|\psi(0)\rangle = |b\rangle$ so $c_k(0) = 0$. Then, we get

$$\dot{b} = -g^2 \sum_{k=-\infty}^{\infty} f^*(t) \int_0^t e^{i\delta_k(t-\tau)} f(\tau) b(\tau) d\tau$$

We are given that f is slowly varying so we approximate $f(\tau) \approx f(t)$. We use the Markovian assumption to do the same for $b(\tau) \approx b(t)$. Then, we get

$$\begin{aligned} \dot{b} &\approx -g^2 |f(t)|^2 b(t) \sum_{k=-\infty}^{\infty} \int_0^t e^{i\delta_k(t-\tau)} d\tau \\ &\approx -g^2 |f(t)|^2 b(t) \sum_{k=-\infty}^{\infty} (\pi \delta(\delta_k) + i\mathcal{P} \frac{1}{\delta_k}) \end{aligned}$$

Now we separate variables to get

$$\frac{db}{b} = -dt g^2 |f(t)|^2 \sum_{k=-\infty}^{\infty} (\pi \delta(\delta_k) + i\mathcal{P} \frac{1}{\delta_k})$$

Integrating and using the initial condition $b(0) = 1 \implies \ln(b(0)) = 0$, we get

$$\ln(b(t)) = \int_0^t d\tau g^2 |f(\tau)|^2 \sum_{k=-\infty}^{\infty} (\pi \delta(\delta_k) + i\mathcal{P} \frac{1}{\delta_k})$$

Thus,

$$b(t) = \exp \left[\sum_{k=-\infty}^{\infty} \left(\pi \delta(\delta_k) + i \mathcal{P} \frac{1}{\delta_k} \right) \int_0^t d\tau g^2 |f(\tau)|^2 \right]$$

Exercise 3b

The outgoing state is spanned by the $|k\rangle$ states, so

$$|\psi^{\text{out}}\rangle = \sum_{k=-\infty}^{\infty} c_k(t) |k\rangle$$

The real-space wavefunction is given by

$$\psi^{\text{out}}(x, t) = \langle x | \psi^{\text{out}} \rangle = \sum_{k=-\infty}^{\infty} c_k(t) \langle x | k \rangle = \sum_{k=-\infty}^{\infty} c_k(t) e^{ikx}$$

With density of states $\frac{dN}{dk} = \rho$, we can take $N = \sum_k \rightarrow \int \rho dk$, so that

$$\begin{aligned} \psi^{\text{out}}(x, t) &= \int_{-\infty}^{\infty} \rho c_k(t) e^{ikx} dk \\ &= ig \int_{-\infty}^{\infty} \rho e^{ikx} \int_0^t e^{i\delta_k(t-\tau)} f(\tau) b(\tau) d\tau dk \\ &= ig\rho \int_0^t f(\tau) b(\tau) \int_{-\infty}^{\infty} e^{ikx} e^{i\delta_k(t-\tau)} dk d\tau \\ &= i2\pi g\rho \int_0^t f(\tau) b(\tau) e^{i\nu(t-\tau)} [\delta(x - v_0(\tau - t)) + \delta(x + v_0(\tau + t))] d\tau \end{aligned}$$